
What Does Your Benchmark Actually Measure? A Training-Free Audit of Held-Out-Form Protocols for Symbolic Expressions

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Abstract

1 A held-out-form benchmark asks a model to recognise that two syntactically
2 different expressions denote the same thing, and accuracy on it is routinely read
3 as evidence that the model has learned algebraic *structure*. That reading is an
4 assumption connecting a protocol to a claim, and it is rarely tested. We test it.
5 On the canonical physics-derived suite every equation carries a variable set no
6 other equation shares, so a *zero-parameter* bag over variable tokens alone is a
7 complete lookup solution by construction, and scores 1.000 where a trained Tree-
8 LSTM scores 0.972: accuracy on this benchmark carries no information about
9 structural learning. We give a four-tier account of what such a benchmark can
10 be solved by (variable identity, operator and arity inventory, order and local tree
11 shape, and genuine composition), release a training-free auditor that measures the
12 lower tiers before any model is trained, and run a targeted audit of 23 corpora
13 from four benchmark families with it. The third tier is the one usually skipped,
14 and it matters: on EQNET poly8 a zero-parameter bag over parent-child and
15 subtree-size statistics reaches 0.411 where the operator bag reaches 0.186, so
16 beating an operator bag establishes far less than it appears to. To hold the lower
17 tiers fixed we pair each held-out form with a *twin* of identical token and operator
18 multisets differing only in how they combine. A rotation twin pins inventory at
19 chance but still leaks shape (our own tree-local bag scores 0.861 on it), so we build
20 a *shape-matched* twin by swapping siblings under a non-commutative operator,
21 which leaves that bag exactly invariant. With inventory and shape both pinned
22 at chance by construction, the untrained encoder falls from 0.925 to 0.788 while
23 the trained one holds at 0.994: the trained-untrained gap *widens* from 0.075 to
24 0.206. That an untrained recursive encoder scores 0.788 at all is why we call these
25 baselines *non-learned* rather than “structure-free”. Pointed at a live leaderboard,
26 the audit yields a non-learned baseline floor under that leaderboard’s own protocol,
27 not a certificate: computing score_5 for a token bag and an untrained encoder
28 on the 14 SemVec corpora, under those authors’ own procedure, no non-learned
29 representation reaches the published systems on any corpus. We report this having
30 first retracted the opposite conclusion, which our own internally-consistent-but-non-
31 matching protocol had produced: a skyline is only a skyline for the evaluation it was
32 computed under. We distil the result into an eight-step checklist, ship step 1 as a
33 command-line tool, and ship a script that recomputes and asserts every non-learned
34 number in this paper.

1 Introduction

Evaluation protocols carry claims. A held-out-form protocol for symbolic expressions (withhold one equivalent form of each equation from training, then ask a learned encoder to identify it) is understood to carry the claim that the encoder has recognised algebraic *structure* rather than surface tokens. The protocol does not establish that claim on its own. Something has to rule out the cheaper explanations, and in this literature that step is usually skipped.

We ran the cheapest possible check on the canonical case. On a 10-equation held-out-form suite built from AI Feynman [Udrescu and Tegmark, 2020], a trained Tree-LSTM identifies held-out forms at 0.972 ± 0.060 ($n=25$ seeds, chance 0.10), against an untrained encoder’s 0.724 ± 0.184 . Read naively, that is structural identification. It cannot be. On this suite every equation has a variable set no other equation shares (§5), so a bag over variable tokens alone (nearest-centroid, *no training and no parameters*) is a complete lookup solution *by construction*, and duly scores 1.000 (sd 0.000), above the trained encoder at every seed. A full-token bag lands nearby but unstably: 0.992 ± 0.027 in this run, 0.932 ± 0.075 in an independent replication of the same protocol, a spread that straddles 0.972 and is why we anchor on the deterministic row (§8). Under this construction the benchmark is solvable by lookup: whatever the encoder learned, this protocol cannot see it, because a method that provably has no notion of tree structure already saturates the metric.

Ten equations is a small suite, and this is a constructed counterexample, not evidence of prevalence: showing that a failure mode exists and can be total needs one case, not a large sample. Prevalence is assessed separately by the 23-corpus audit (§5) and behaviour at scale by a published leaderboard (§6). With $K=10$ the unit of inference also matters from the outset: per-equation leakage ranges from -0.05 to 1.12 , so the $\pm$ figures above are seed-level and understate equation-level uncertainty by roughly $6\times$; we bootstrap over equations throughout (§8).

This gives the paper its thesis. *Held-out-form accuracy can be explained by progressively stronger non-learned cues; we give a training-free audit and a controlled-probe protocol for determining which level of explanation a benchmark actually rules out.* The failure mode throughout is not a model that underperforms but a measurement whose connection to the claim it supports was never checked.

A tier decomposition of what a benchmark can be solved by. “Structure” is not one thing. We separate variable identity (tier 1), operator and arity inventory (tier 2), order and local tree shape (tier 3), and genuine composition (tier 4), and argue that identification claims must name their tier (§2). The distinction is not cosmetic: a benchmark can be fully immune at tier 1 and still be separable at tier 2, which is the situation almost everywhere we looked, and clearing tier 2 still leaves tier 3, where surface statistics of a tree substitute for composing it.

Testing for what the lower tiers cannot explain. We measure tier 3 with two zero-parameter baselines (token n-grams; parent–child and subtree-size statistics), then hold both inventory *and* local shape fixed with a *twin* probe: each held-out form paired with a rearrangement of itself carrying identical token and operator multisets, so those baselines are pinned at exactly chance by construction (§3). This tests sensitivity to compositional distinctions not captured by the measured tier-3 statistics; it does not isolate composition, and §3 says why no such probe could. The trained encoder is near ceiling there, but an *untrained* encoder reaches 0.788, which is why we call these baselines *non-learned* rather than “structure-free”: a randomly initialised recursive encoder computes over structure without having learned any.

A training-free screen, and a targeted audit of 23 corpora. Tier separability is computable from the corpus alone, with no model and no training run (§4). We ship it as `audit-symbolic-benchmark` and apply it to 23 published corpora across four families (§5). The result narrows the cautionary claim considerably: the tier-1 flaw is a property of a per-symbol *tokenization convention* applied to physics corpora, not of equivalence-class benchmarks in general. The canonical published family, EQNET [Allamanis et al., 2017] and successors [Gangwar and Kani, 2023, Zheng et al., 2025], defines classes over a shared variable pool and is immune at tier 1 ($\leq 3.2\%$ unique variable sets across all 15 corpora). Tier 2 is where the exposure is.

stage	establishes	does <i>not</i> establish
1. corpus screen	a lower-tier cue <i>may</i> exist	that it is realised
2. matched baseline	whether the cue <i>realises</i> under the same protocol as the claim	which cue the model uses
3. controlled probe	sensitivity beyond the <i>tested</i> lower-tier cues	composition
trained vs. untrained	what is attributable to training <i>relative to the architecture</i>	a learned/architectural split

no stage establishes that the encoder performs compositional algebraic reasoning, see §3

Figure 1: **What each stage licenses.** The tools form a pipeline, and conflating its stages is what produces overclaims. A claim is worth exactly the highest floor it clears: the screen is a cheap pre-training diagnostic that says where to look, only the matched baseline licenses a statement about a *benchmark*, and only the controlled probe licenses one about a *model*. The two non-learned families answer different questions and are reported separately throughout: a **bag** is a structure-free lower-tier floor, an **untrained encoder** is an architecture-sensitive non-learned floor that already computes over structure.

86 **A floor under a live leaderboard, and a retraction.** The strongest test of an audit is to point
87 it at numbers people are currently standing behind. Three published systems report score₅ on the
88 UNSEENEQCLASS split of the 14 SemVec corpora. The metric needs only a distance, so it is
89 computable for a token bag exactly as for a learned embedding. Computed under those authors’
90 own procedure, the audit yields a baseline floor under the published protocol: no non-learned
91 representation reaches EqNet or the headline SemEmb system on any of the 14 corpora (§6). We
92 report this having first published a stronger claim internally and retracted it: our own protocol,
93 internally consistent and satisfying our own checklist, made the same floor appear to beat a published
94 ablation on 13 of 14 corpora instead of 5. Every difference from the original procedure had pushed
95 our numbers up. *A skyline is only a skyline for the evaluation it was computed under.*

96 **What the audit costs the auditor.** We applied the standard to ourselves and it was not free:
97 supplying a baseline we had not previously measured repeatedly reversed conclusions we had drawn,
98 and two purpose-built “leakage-proof” suites turned out to be solvable at tier 2 (§8). The unit of
99 inference also matters: per-equation leakage ranges from -0.05 to 1.12 , so a seed-level interval
100 understates equation-level uncertainty by roughly $6\times$.

101 We close with the eight-step checklist these findings imply (§9).

102 **Related work.** None of the machinery is new in kind. Geirhos et al. [2020] frame *shortcut learning*
103 as a pervasive failure mode; in NLI, hypothesis-only baselines matched full models and exposed
104 annotation artifacts rather than reasoning [Gururangan et al., 2018, Poliak et al., 2018]. Our zero-
105 parameter bag is the symbolic analogue, and on the physics suite it does not merely match the
106 trained encoder but solves the benchmark by construction. Hewitt and Liang [2019] established with
107 *control tasks* that probing accuracy is interpretable only against a randomised baseline, which our
108 untrained-encoder skyline supplies. Renaming-invariance has an analogue in code: Type-2 clones
109 are by definition identical up to identifier names, and clone benchmarks curate them as a distinct
110 category [Svajlenko et al., 2014]. SCAN [Lake and Baroni, 2018] and COGS [Kim and Linzen, 2020]
111 showed that how a held-out set is *constructed* decides observed difficulty, and the contamination
112 literature [Jacovi et al., 2023, Zhou et al., 2023] recommends shortcut-breaking held-out sets, which
113 our checklist operationalises as a tool. What is new is the tier decomposition, the training-free screen
114 behind it, and the audit of which families are exposed at which tier.

115 2 A Hierarchy of Cues

116 A benchmark can be solvable at several levels, and those levels deserve different verdicts. We fix a
117 four-tier hierarchy and use it consistently, including when judging our own results.

118 **Tier 1: variable identity.** Which variable symbols appear. Purely notational: under a consistent
 119 relabelling the expression denotes the same function, so any identification that depends on it is a
 120 shortcut, always.
 121 **Tier 2: operator and arity inventory.** Which operators appear, how many, and over how many
 122 variables. Semantically meaningful, x_0/x_1 really is a different function from $x_0 \cdot x_1$, but still
 123 bag-level: a model reading only the multiset has not parsed anything.
 124 **Tier 3: order, depth and local shape.** Token adjacency, depth profile, parent–child operator statis-
 125 tics, subtree sizes. $x_0 + x_1 \cdot x_2$ and $(x_0 + x_1) \cdot x_2$ differ in all of these. Crucially, *none of it*
 126 *requires composition*: these are surface statistics of a tree, computable without recursion.
 127 **Tier 4: compositional structure.** How the operators actually combine.

128 **Why tier 3 has to exist.** An earlier version of this hierarchy went straight from tier 2 to composition,
 129 licensing a bad inference: *encoder beats a bag* $\Rightarrow$ *encoder learned structure*. Everything in tier 3
 130 is available to a model that never composes anything, so clearing a bag establishes only that the
 131 representation uses information beyond the tested bag cues. §3 measures that gap.

132 **Non-learned is not structure-free.** Our *bag classifiers* (variable-only, variable-blind, full-token,
 133 and the tier-3 baggers) are nearest-centroid over a multiset with no parameters. The *untrained*
 134 *encoder* is the trained architecture, randomly initialised. Both are *non-learned*, but only the bags
 135 are structure-free: a random Tree-LSTM still has recursive composition, parent/child interaction
 136 and depth sensitivity, computing over structure without having learned any. Hence **non-learned**
 137 **baselines**, not “structure-free” ones, a distinction §3 measures rather than stipulates. Claims must
 138 accordingly name their tier: we report identification *above the strongest non-learned baseline*, and
 139 reserve claims about composition for the twin probe.

140 2.1 Setup

141 Expressions are binary trees over $\{+, -, *, /, \sin, \cos, \exp, \log, \sqrt{\cdot}\}$. The physics suite is the 10
 142 AI Feynman equations [Udrescu and Tegmark, 2020] with 1–3 variables, each mapped to a distinct
 143 variable subset of a 16-symbol pool; that mapping is the convention under audit. Equivalence classes
 144 come from a 13-schema paraphrase library and have ≈ 3 members differing by a single rewrite, the
 145 easiest possible positive, which bounds every number we report. Encoders are Tree-LSTMs [Tai et al.,
 146 2015], GINs [Xu et al., 2019] and Transformers [Vaswani et al., 2017], 128-dimensional, trained
 147 with satisfaction classification plus a semi-hard triplet loss [Schroff et al., 2015]. The metric is the
 148 fraction of held-out embeddings whose nearest training-class centroid is correct; chance $1/K$ is a
 149 uniform-class conceptual baseline, not the empirical expectation under every retrieval construction.
 150 Every baseline is evaluated under the identical centroid protocol, which is what makes it comparable
 151 to the trained number rather than merely adjacent to it.

152 **Neither baseline substitutes for the other.** On EQNET ONEVARPOLY13 the bag scores 0.15
 153 where the untrained encoder scores 0.66; on Feynman-10 the ordering reverses (1.00 versus 0.13). A
 154 recursive nonlinearity is sensitive to shape a bag discards, and blind to counts a bag sees, so both are
 155 reported, always.

156 3 Measuring the Gap: Order, Shape, and What Is Left

157 Tier 3 is only a useful category if it can be measured, and a claim above it is only worth as much as
 158 the cues that were held fixed while testing it. We do both, and each attempt corrects the last.

159 **Two tier-3 baggers (tag r71).** *Token n-gram* takes bigrams and trigrams over a bracketed depth-first
 160 linearisation (unigrams excluded, so it measures adjacency rather than inventory); *tree-local* takes
 161 parent→child pairs, depth-indexed labels and a subtree-size histogram. Both are zero-parameter and
 162 *variable-blind*, since tier-1 power is already measured by the variable-only bag. They matter: on
 163 poly8 unseen classes the tree-local bag reaches 0.411 at $K=500$ against the variable-blind bag’s
 164 0.186, so over half the headroom one might attribute to “structure beyond operator content” is local
 165 shape.

Table 1: **Three controlled probes, across six corpora** (tag r74; 3 seeds; chance 0.500; ranges over corpora). Each probe pairs every held-out form with a twin of identical token and operator multisets; they differ in *which* lower-tier cues they additionally hold fixed. **Bold** entries are pinned at exactly 0.500 *by construction*, verified per instance rather than measured. Reading down: rotation leaves shape free and the shape bag exploits it (0.365–0.909); the swap closes shape but leaves order; relocation closes order too. The trained–untrained gap survives all three, so the residual is not argument order. Per-cell values, coverage, and the skipped and bags-only cells are in Appendix B.

Probe	<i>non-learned baselines</i>			<i>encoder</i>		gap	corpora
	inventory	shape	order	untrained	trained		
Rotation	.500	.365–.909	.424–.775	.586–.940	.975–1.000	+.060–+.414	5
Swap	.500	.500	.436–.769	.748–.897	1.000	+.103–+.252	5
Relocation	.500	.500	.500	.617–.823	.965–1.000	+.142–+.383	4

A ladder of controls, each closing the previous one’s loophole. We pair every held-out form with a *twin* carrying identical token and operator multisets and ask whether the encoder places the true form closer to its class centroid (chance 0.5). Table 1 summarises the three transformations, in increasing strictness; per-corpus detail is in Appendix B and the earlier two-probe comparison on poly8 in Appendix E.

Rotation (r72) sends $(a \text{ op}_1 b) \text{ op}_2 c \rightarrow a \text{ op}_1 (b \text{ op}_2 c)$: inventory is pinned, but the rotation moves depths, parent–child pairs and subtree sizes (exactly tier 3) and our own measurement exposes the leak, tree-local reaching 0.832 on poly8 and 0.909 on poly5. *Sibling swap* (r73) sends $x - y \rightarrow y - x$ under a non-commutative parent; since the tree-local bagger does not distinguish left child from right, its bag is left *exactly* invariant, pinning shape as well as inventory but not order. *Operand relocation* (r74) exchanges two leaves holding *different* variables, $(x_0 - x_1) \cdot x_2 \rightarrow (x_0 - x_2) \cdot x_1$; the tier-3 baggers being variable-blind, inventory, shape *and* order are all invariant, leaving only the variable-to-position binding. Every twin is verified per instance: matching multisets, matching bags where promised, and a numerical (for boolean, exhaustive truth-table) check that it denotes a different function.

The residual is not argument order. A reasonable objection to the swap is that it tests sensitivity to left/right ordering rather than anything more general. Relocation answers it directly: there the n-gram baseline is pinned at 0.500 by construction, and the trained–untrained gap persists on all four corpora where relocation exists (+0.142, +0.383, +0.210, +0.184), in the same range as the swap. Across 17 cells spanning six corpora and two symbolic families, every construction guarantee held and every encoder gap was positive.

What this licenses, stated exactly. Not “the encoder composes”. A sufficiently rich set of local relations determines a tree, so tiers 3 and 4 are separated by the *boundedness* of the statistics we chose (capped depth, capped subtree size, bags of local pairs), not categorically. The honest reading is falsification-shaped: the encoder survives every non-learned explanation we could construct, including one fixing inventory, shape and order at once, and a claim is worth exactly the highest floor it clears. The gap likewise bounds what training can be credited with *under this probe* without decomposing into architectural and learned parts, which interact. That an *untrained* recursive encoder already scores 0.586–0.940 here is why we call these baselines *non-learned*, not “structure-free”.

4 The Auditor: Stage One of Three

Figure 1 states the pipeline and, for each stage, what it does and does not license; conflating those stages is what produces overclaims. This section is stage 1.

Tiers 1–3 are properties of the corpus, so they can be screened with no model, no GPU and no forward pass. Given a corpus of equivalence classes the auditor reports, per tier, whether class representatives are separable by that tier’s features alone: the *unique var-set fraction* (tier 1: if high, knowing which variables appear pins the class before any model exists), *operator-bag separability* with its collision profile (tier 2), the *arity distribution* (which often partitions a small suite into near-singleton groups, a confound aggregate accuracy hides), and the tier-3 statistics of §3. Figure 2 (Appendix A) shows it

on two corpora; the tool is dependency-light and takes a documented JSON format, with adapters for every corpus in Table 3.

What it does not do. These columns measure whether class *representatives* differ; they do not predict what a baseline scores on *held-out forms*. Across 12 corpora no zero-parameter separability statistic we tried predicts untrained-encoder accuracy ($R^2 = -2.8$; Appendix H), reported as a limitation rather than tuned until it correlated. Separability is a *diagnostic*; measured baseline performance is the *evidentiary* criterion. A cheap pre-training screen that tells you where to look is worth having even when it cannot tell you how much the cue will be worth.

5 Where Do Published Benchmarks Expose Low-Tier Cues?

Before diagnosing any encoder, we ask whether lower-tier separability is a property of the *corpora* the field uses. Table 3 (Appendix C) applies the auditor to 23 corpora from four benchmark families, a targeted audit, not a comprehensive survey. Nothing here is trained and nothing is fitted.

The tier-1 flaw is real, and it is confined to one family. AI Feynman [Udrescu and Tegmark, 2020] gives 84% of its equations a variable set no other equation shares, so a protocol built on it is largely solvable without algebraic reasoning, as §1 showed directly. But that is a property of a *tokenization convention*: physics equations are written with named quantities, and giving each its own symbol index makes the variable set a class fingerprint. It is not a property of equivalence-class benchmarks generally. Generated corpora (NeSymReS [Biggio et al., 2021], Lample–Charton [Lample and Charton, 2020]) reuse a tiny pool and score 0%; so, decisively, does **EQNET** [Allamanis et al., 2017], at most 3.2% across all 15 of its corpora: there variable identity is *semantic* ($a - b$ and $b - a$ are different classes), so there is no tier-1 shortcut to remove.

We state this narrowly because the survey supports nothing broader: we cannot cite published work running this protocol on a physics corpus and attributing the result to structural learning. The defensible claim is design hygiene, and the screen says so in advance. This is a targeted audit of selected published corpora, not a comprehensive survey.

Tier 2 is the general exposure, and it is not a cheat. Read the last column instead: operator-bag separability is 66–100% across every family, including corpora perfectly clean at tier 1. Unlike tier 1, this is not illegitimate information. Distinguishing $x_0 + x_1$ from x_0/x_1 by operator inventory is *meaningful*: they are different functions, and a model that notices is not cheating. The problem is only ever a mismatch between cue and claim: a corpus at 0% tier 1 and 92% tier 2 supports “the encoder distinguishes these classes” and not “the encoder parses these expressions”. That is the finding we would most like taken away, because the community has largely closed tier 1 by adopting shared variable pools while leaving tier 2 open, and §3 shows tier 3 sits between even those.

6 A Floor Under a Published Leaderboard

The strongest form of the audit is to point it at numbers someone is currently standing behind. Three published systems report score_5 on the UNSEENEQCLASS split of the 14 SemVec corpora: EqNet [Allamanis et al., 2017], and the StructEmb ablation and headline SemEmb system of Gangwar and Kani [2023]. The metric needs only a distance, so it is computable for a bag exactly as for a learned embedding. Table 5 (Appendix F) does so for a token bag, a variable-blind bag and an untrained Tree-LSTM, evaluated exactly as Allamanis et al. [2017] do: neighbours searched in the *full corpus* rather than the test split, and normalisation by $\min(|\text{class}|, 5)$.

The leaderboard is robust. No non-learned representation reaches EqNet or SemEmb on *any* of the 14 corpora, usually by a wide margin (BOOL8: bag 12.7 against EqNet’s 58.1). Five corpora see one reach the StructEmb ablation, the weakest published column. The verdict is a *non-learned baseline floor under the published evaluation protocol*, deliberately weaker than a certificate: as §3 shows, exceeding a bag does not by itself establish that the information used is compositional.

7 Protocol Mismatch Manufactures Evidence: A Retraction

Our first attempt at this comparison produced the opposite result, and the way it failed is the most transferable finding in this paper.

That attempt computed score_5 with the UNSEENEQCLASS split serving as its own neighbour pool, and normalised by k . Under that procedure the non-learned floor looked dramatic: a bag or an untrained encoder matched or beat the StructEmb ablation on 13 of 14 corpora, beat the headline SemEmb system on 3, and reached 69.4 on BOOL8 where EqNet reports 58.1. Had we stopped there, we would have published a claim that a zero-parameter baseline undercuts a published leaderboard.

It was wrong, and the direction of the error was not random. Three differences from the published procedure, neighbour pool, normalisation, and query construction (Appendix G), *each* pushed our numbers up. On BOOL8 the published protocol searches 257,784 candidate neighbours where ours searched 13,859, and more distractors can only lower a score. Reproducing all three moves the comparison from “matches or beats StructEmb on 13 of 14” to 5 of 14, and from “beats SemEmb on 3” to none.

Why this is not just our bug. We had done what a careful auditor is supposed to do. We reported *both* non-learned baselines, under a *single, internally consistent* protocol, satisfying our own checklist, and still produced a badly misleading comparison, because the protocol was ours and the numbers beside it were someone else’s. Internal consistency is not the relevant property when a baseline is placed in a table next to published results; what matters is whether the baseline was computed under *the same evaluation* as the numbers it is being compared against. Every non-learned floor, hypothesis-only baseline, or majority-class control reported against an external leaderboard is exposed to this error, and the error has a direction: the shortcuts that make a protocol convenient to re-implement (evaluate within the test split, normalise by k) are the ones that inflate the floor.

We keep the superseded numbers on record rather than deleting them, and we add a checklist step for it (step 7, §9): verify the evaluation matches what you compare against.

8 Reporting Practice: Units of Inference and Sweeping for the Class

Two further findings concern how results are reported rather than what they are, and both cost us claims.

The unit of inference is equations, not seeds. A seed-level interval answers “would a rerun give this number?” when the question of interest is “would other equations give this number?” The two differ substantially: per-equation leakage on the 10-equation suite ranges from -0.05 to 1.12 , so resampling equations rather than seeds widens the interval by roughly $6\times$ (Appendix I). We report equation-level bootstraps throughout.

Fix the class, not the instance. Four times in this work, supplying a non-learned baseline we had not previously measured reversed a conclusion we had drawn: on a purpose-built 12-law suite, on a 5-law suite, on the score_5 leaderboard (§7), and on the interpretation of a protocol-sensitivity contrast. Each time we had fixed the instance in front of us without sweeping for the class.

The two suites are worth naming, because we built them specifically to be “leakage-proof” and the audit says they are, at the wrong tier. Both appear in the last block of Table 3 at exactly 0% unique variable sets: every law is defined over the same shared variables, so the variable-identity shortcut we set out to eliminate is genuinely gone. Both also measure exactly 100% operator-bag separability at tier 2. On the 5-law suite under the joint objective, the consequence is unambiguous: an *untrained* encoder scores 0.40 against the trained Tree-LSTM’s 0.32 ± 0.098 ($n=25$, chance 0.20); two of the five laws are operator-homogeneous and are separated under any projection, so every random seed scores exactly $2/5$. We had removed tier 1 and declared victory without measuring tier 2.

The remedy is mechanical. For *every* protocol under which we state an identification number we measure all three bag representations and an untrained encoder *under that same protocol*, at $n=25$ seeds (Appendix D).

298 The verdicts split cleanly. On the three physics-derived protocols a zero-parameter bag over variable
 299 tokens alone is perfect and the trained encoder is *below* it, so those protocols cannot license a claim
 300 about structure, this paper’s thesis applied to its own measurements. Only the shared-variable design
 301 and the two EQNET corpora put the trained encoder above every non-learned baseline, and those are
 302 the only identification claims we advance. Knowing which rows survive is possible only because the
 303 sweep was run for every row, not for the rows we doubted.

304 One row audits the sweep itself. The Feynman E3 full-token bag reads 0.932 here against the 0.992
 305 of §1. These are *not* different protocols: the sweep re-runs it exactly, and the two disagree only
 306 because the held-out paraphrase is redrawn. The estimates are within one standard deviation, but they
 307 straddle the trained encoder’s 0.972, so a comparison resting on that margin reverses between two
 308 runs of the same experiment. Hence the claim in §1 is stated against the variable-only bag: 1.000
 309 with sd 0.000, perfect by construction rather than an estimate that happened to land high.

310 9 The Checklist

311 If only one sentence survives from this paper, let it be this. **Before interpreting an identification**
 312 **score as structural learning, report matched non-learned baselines for every claimed evaluation**
 313 **protocol, and verify the benchmark cannot be solved by lower-tier cues.** The eight steps below
 314 are its operational expansion; steps 1–2 would have caught the failure we opened with, and step 7
 315 would have caught our own retraction.

- 316 1. **Screen the corpus before training.** A high unique var-set fraction means a tier-1 lookup table,
 317 and no training result on it will mean what you want it to.
- 318 2. **Report both non-learned baselines under the identical protocol:** a bag classifier and a
 319 randomly initialised encoder. Neither substitutes for the other (§2).
- 320 3. **Run within-range variable renaming,** remapping into the *trained* index range, and report the
 321 (plain, renamed) pair rather than a summary index.
- 322 4. **Audit positive results at tiers 2 and 3:** operator-scramble and arity controls, plus the order and
 323 local-shape baselines of §3. Clearing a bag is not enough.
- 324 5. **Verify the control is an isomorphism of the benchmark.** On a de Bruijn-normalised corpus the
 325 naive in-range remap is the identity, and the control silently does nothing.
- 326 6. **Verify held-out forms are genuinely out-of-library:** not string-identical to, or one library
 327 rewrite from, a training form.
- 328 7. **Verify the evaluation matches what you compare against:** same neighbour pool, normalisation
 329 and query construction as the published numbers (§7).
- 330 8. **Make the population the unit of inference.** Bootstrap over classes, not seeds.

331 Together they ask one question: does this benchmark license the claimed level of interpretation? It is
 332 a screen, not a verdict of impossibility: clean corpora exist (§5), and §3 clears the bar on one.

333 **What we claim, exactly.** The hierarchy is a *falsification* device, not an identification one: failing
 334 a tier invalidates the stronger reading, while clearing one establishes only insensitivity to the cues
 335 actually tested. Applied to our own results:

336 **Established:** the physics benchmark admits a complete tier-1 shortcut. **Established:** the trained
 337 encoder exceeds every non-learned baseline we could construct, including one holding inventory, local
 338 shape *and* order fixed by construction. **Not established:** that the encoder performs compositional
 339 algebraic reasoning. That last line is not modesty: no probe can partition “composition” from “a
 340 structural statistic we did not measure”, since a sufficiently rich set of local relations determines the
 341 tree. A claim is worth exactly the highest floor it clears.

342 **Reproducibility and limitations.** A paper arguing that protocol details decide conclusions should
 343 not ask to be taken on trust. Screen, probes, run tags and `verify_claims.py` (which recomputes
 344 every zero-parameter number here and *asserts* it, exiting non-zero on disagreement) are in the
 345 supplementary material; both leaderboards ship as tags, so §7 reproduces in the order it happened.
 346 Limits: our corpora are algebraic and boolean, the screen is untested elsewhere, and near-degenerate
 347 class sizes (§2.1) bound every result.

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399 A The Screen, on Two Corpora

400 Stage 1 of the auditor runs in seconds and trains nothing. Figure 2 reproduces its output on two
 401 representative corpora: the physics corpus feynman100, where it fires the tier-1 warning that
 402 motivates this paper, and EQNET poly8, where it reports tier-1 clean and then flags that pairwise
 403 separability far exceeds the measured bag accuracy.

```
$ audit-symbolic-benchmark --builtin feynman100
tier 1 unique var-set 84.0% tier 2 op-bag sep. 98.9%
WARNING: a majority of classes have a unique variable set. A held-out-form
protocol on this corpus is largely solvable by token lookup; report a renaming
control before attributing identification to learned structure.

$ audit-symbolic-benchmark --builtin eqnet:poly8
tier 1 unique var-set 0.0% tier 2 op-bag sep. 82.2%
MEASURED bag accuracy 0.148
NOTE: pairwise separability is far above the measured bag accuracy (98% vs 15%).
Separability says the classes *differ*; accuracy says whether a bag can actually
place a held-out form. Report the measured number.
```

Figure 2: **Stage 1 of the pipeline, on two corpora** (abridged real output; runs in seconds, no training). On the physics corpus it fires the tier-1 warning that motivates this paper. On poly8 it reports tier-1 clean and then, unprompted, warns that pairwise separability far exceeds what a bag actually *scores*: the tool enforcing the distinction of §4 rather than leaving it to the reader.

404 B The Controlled-Probe Matrix, Every Cell

405 This is the full controlled-probe matrix underlying §3. Figure 3 shows what each baseline can solve
 406 on the rotation and swap twins, and Table 2 reports every cell of the three-transformation ladder
 407 (rotate, swap, relocate) across all six corpora.

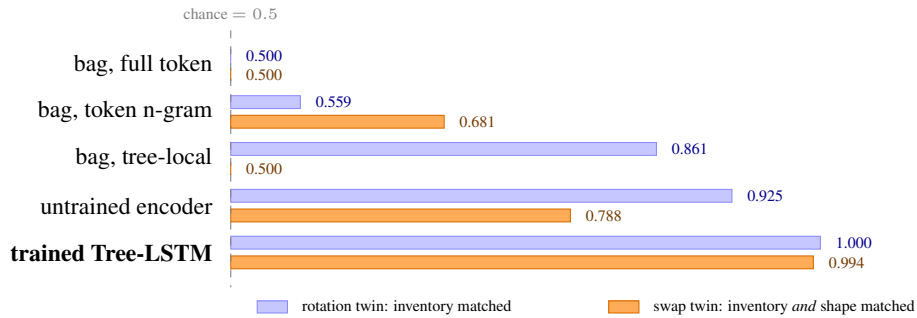


Figure 3: **What can solve the twin probe, and what happens when you close the shape loophole.** Each held-out form on EQNET poly8 is paired with a twin of identical token and operator multisets ($K=500$; chance 0.5; tags r_{72} , r_{73}). Under the *rotation* twin (blue) inventory is matched but local shape is not, and the tree-local bag scores 0.861, so that probe cannot attribute the encoder’s margin to composition. Under the *swap* twin (orange) a sibling exchange leaves the tree-local bag *exactly* invariant, so both inventory and shape are pinned at 0.500 by construction. Closing the loophole costs the untrained encoder $0.925 \rightarrow 0.788$ while the trained encoder holds at 0.994: the trained–untrained gap *widens* from 0.075 to 0.206.

Table 2: **The controlled-probe matrix, every cell** (tag r74; 3 seeds; chance 0.500; $K \leq 150$). Each row pairs every held-out form with a twin of *identical* token and operator multisets. The three transformations form a ladder: rotation matches inventory but perturbs local shape; the sibling swap matches shape too but changes order; **relocation matches inventory, shape and order**, leaving only the variable-to-position binding. Values in **bold** are pinned at exactly 0.500 *by construction*: verified per instance, not measured. The trained-untrained gap survives on every corpus under the strictest probe, so the residual is not argument order. *Skipped and bags-only cells are shown, not omitted*: relocation needs two distinct variables, which ONEVARPOLY13 lacks; BOOLEAN8 has no encoder row because the encoder vocabulary contains no boolean operators. n is twins constructed (coverage in parentheses).

Corpus	Probe	n (cov.)	<i>non-learned baselines</i>			<i>encoder</i>		gap
			full	shape	order	untr.	trained	
POLY8	rotate	116 (.77)	.500	.832	.569	.940	1.000	+.060
	swap	104 (.69)	.500	.500	.625	.808	1.000	+.192
	relocate	113 (.75)	.500	.500	.500	.823	.965	+.142
SIMPLEPOLY8	rotate	37 (.39)	.500	.365	.568	.586	1.000	+.414
	swap	70 (.74)	.500	.500	.436	.748	1.000	+.252
	relocate	54 (.57)	.500	.500	.500	.617	1.000	+.383
SIMPLEPOLY10	rotate	57 (.38)	.500	.570	.456	.637	1.000	+.363
	swap	116 (.77)	.500	.500	.634	.779	1.000	+.221
	relocate	84 (.56)	.500	.500	.500	.790	1.000	+.210
POLY5	rotate	66 (.60)	.500	.909	.424	.838	1.000	+.162
	swap	58 (.53)	.500	.500	.509	.897	1.000	+.103
	relocate	69 (.63)	.500	.500	.500	.802	.986	+.184
ONEVARPOLY13	rotate	134 (.89)	.500	.881	.545	.751	.975	+.224
	swap	106 (.71)	.500	.500	.769	.846	1.000	+.154
	relocate	<i>skipped: one variable, so no relocation exists</i>						
BOOLEAN8	rotate	71 (.47)	.500	.831	.775	—	—	—
	swap	52 (.35)	.500	.500	.721	—	—	—
	relocate	111 (.74)	.500	.500	.500	—	—	—

408 C The Targeted Corpus Audit, Full Table

409 Table 3 gives the complete corpus audit summarised in §5: all 23 corpora from the four benchmark
410 families, with their tier-1 (unique variable-set) and tier-2 (operator-bag separability) figures.

Table 3: **Targeted audit of 23 corpora from four benchmark families** (no training, nothing fitted; tag r57). *Unique var-set* (tier 1): fraction of classes whose variable *set* no other class shares; if high, the protocol is solvable by token lookup. *Op-bag sep.* (tier 2): fraction of class pairs separable by the operator/arity bag alone. **The tier-1 failure is confined to the physics block**; the canonical published family, EQNET [Allamanis et al., 2017], is immune ($\leq 3.2\%$ across all 15 corpora), as are generated corpora. **Tier 2 is high everywhere, including where tier 1 is clean**, which is the generalizable finding. These columns measure whether class *representatives* differ, not what a bag scores on held-out forms; the two diverge (Appendix H). N is the class count; corpora above 600 classes are subsampled (seeded) for the $O(n^2)$ pairwise columns.

Benchmark	N	Var. pool	Unique var-set (T1)	Op-bag sep. (T2)
<i>Physics-derived held-out-form benchmarks (one variable subset per equation):</i>				
AI Feynman [Udrescu and Tegmark, 2020]	100	93	84%	99%
Feynman-10 (this work)	10	15	100%	98%
Synthetic 8-law (this work)	8	11	100%	96%
<i>EQNET / semvec [Allamanis et al., 2017]: the published equivalence-class family:</i>				
POLY (7 corpora)	47–1102	1–3	0%	66–92%
BOOL (8 corpora)	95–7312	3–10	0–3%	69–97%
<i>Generated SR / integration corpora (shared variable pool):</i>				
NeSymReS [Biggio et al., 2021]	200	3	0%	100%
Lample–Charton [Lample and Charton, 2020]	320	1	0%	99%
<i>Shared-variable designs (this work, the fix):</i>				
Feynman-10 shared (de Bruijn)	10	4	20%	98%
Overlap-5 (shared x_0, x_1)	5	2	0%	100%
Overlap-12 (shared x_0, x_1)	12	2	0%	100%

D The Full Non-Learned Baseline Sweep

For every protocol under which the paper states an identification number, all three bag representations and an untrained encoder, measured under that same protocol ($n=25$ seeds):

Protocol	K	strongest non-learned	trained	verdict
Matched contrast, in-library	8	1.000 (var-only)	0.965	no claim
Matched contrast, out-of-library	8	1.000 (var-only)	0.832	no claim
Feynman E3 (10 equations)	10	1.000 (var-only)	0.972	no claim
Shared-variable Feynman	10	0.628 (untrained)	0.932	survives
EQNET SIMPLEPOLY5	38	0.368 (full-token)	0.944	survives
EQNET SIMPLEPOLY8	90	0.467 (full-token)	1.000	survives

E Twin Probes: Full Numbers

Table 4 gives the full twin-probe numbers for every baseline under both the rotation twin (inventory matched) and the shape-matched swap twin (inventory *and* local shape matched).

Table 4: **Two twin probes, and what closing the shape loophole does** (EQNET poly8, $K=500$, 3 seeds, chance 0.500; tags r72, r73). Both twins have token and operator multisets identical to the original, so inventory baselines are pinned at exactly 0.500 by construction. The *rotation* twin additionally changes depths, parent-child pairs and subtree sizes (tier-3 cues), which is why the tree-local bag reaches 0.861 there. The *swap* twin exchanges two siblings under a non-commutative operator, leaving the tree-local bag exactly invariant, so shape is pinned too. Under the stricter probe the untrained encoder loses most of its advantage while the trained encoder does not.

Baseline	Cue it can still see	Rotation	Swap
Bag, full token (0 params)	— (pinned)	0.500	0.500
Bag, tree-local (0 params)	shape (rotation only)	0.861	0.500
Bag, token n-gram (0 params)	order	0.559	0.681
Untrained encoder (non-learned)	order, shape, recursion	0.925	0.788
Trained Tree-LSTM	learned	1.000	0.994
<i>margin over the untrained architecture</i>		0.075	0.206
<i>classes probed</i>		395/500	348/500

418 F The score₅ Leaderboard Floor, in Full

419 Table 5 gives the complete score₅ leaderboard floor: all 14 SemVec corpora, our three non-learned
420 representations, and the three published systems, computed under those authors’ own protocol.

Table 5: **A zero-parameter floor under a published leaderboard, computed under that leaderboard’s own protocol** (tag r68). score₅ (%) on the UNSEENEQCLASS split of the 14 SemVec corpora. The right-hand columns are transcribed from the published papers: EqNet [Allamanis et al., 2017], and the StructEmb autoencoder ablation and headline SemEmb system of Gangwar and Kani [2023]. The left-hand columns are non-learned representations we compute: a token-multiset bag, a variable-blind bag (operators, constants and arity), and an untrained Tree-LSTM ($n=3$ initialisations, over the full pool in every case). Nothing on the left is trained. Crucially, all of it is evaluated exactly as Allamanis et al. [2017] evaluate: neighbours are searched in the *full corpus (pool)*, not in the test split; cosine distance; self excluded; and the score normalised by $\min(|\text{class}|, 5)$ rather than by 5. **No non-learned representation reaches EqNet or SemEmb on any corpus.** Boldface marks the five where one reaches the StructEmb ablation.

Corpus	Pool	Queries	<i>non-learned (this work)</i>			<i>published</i>		
			Bag	Var-blind	Untrained	EqNet	StructEmb	SemEmb
Bool8	257784	13859	12.7	2.6	31.6	58.1	30.6	100.0
oneV-Poly13	107725	8032	3.7	2.6	32.9	90.4	38.7	99.6
SimpPoly10	57909	6189	31.8	8.2	36.1	99.3	40.1	99.8
SimpBool8	39048	3096	9.5	5.7	44.5	97.4	36.9	99.4
Bool10	51200	8600	4.8	2.1	14.4	71.4	10.8	91.3
SimpBool10	26304	4536	6.6	2.0	37.9	99.1	24.4	95.5
BoolL5	36050	4656	30.3	0.2	22.3	75.2	38.3	36.0
Poly8	11451	1234	10.6	0.6	26.2	86.2	32.7	87.3
SimpPoly8	3477	362	38.5	2.2	31.9	98.9	47.6	98.9
SimpBoolL5	10050	1242	51.5	0.0	23.1	85.0	55.1	71.1
oneV-Poly10	1291	102	44.1	27.5	47.2	81.3	59.8	74.1
Bool5	1239	154	34.7	11.6	22.9	65.8	36.4	57.7
Poly5	516	74	41.8	7.0	31.5	55.3	5.7	45.2
SimpPoly5	237	24	9.4	0.0	11.1	65.6	28.1	20.8

421 G The Superseded score₅ Comparison, in Full

422 Our first attempt at the leaderboard comparison of §6 computed score₅ with the UNSEENEQCLASS
423 split serving as its own neighbour pool, and normalised by k . Under that procedure a bag or an
424 untrained encoder matched or beat the StructEmb ablation on 13 of 14 corpora, beat the headline

425 SemEmb system on 3, and reached 69.4 on BOOL8 where EqNet reports 58.1. Every difference from
426 the published procedure pushed our numbers up:

- 427 1. *Neighbour pool*. Allamanis et al. [2017] embed the entire corpus and search neighbours in it;
428 the test split only supplies queries. On BOOL8 that is 257,784 candidate neighbours rather than
429 13,859. More distractors can only lower a score.
- 430 2. *Normalisation*. Their score divides by $\min(|\text{class}|, k)$, the number of same-class neighbours
431 actually achievable, not by k .
- 432 3. *Query set*. Their queries are corpus entries whose token string appears in the test split, so each
433 query sits in the pool alongside the rest of its class.

434 Reproducing all three moves the comparison from “matches or beats StructEmb on 13 of 14” to 5 of
435 14, and from “beats SemEmb on 3” to none. We keep the superseded numbers on record rather than
436 deleting them.

437 H What the Separability Screen Does Not Predict

438 The auditor’s separability columns measure whether class *representatives* differ; they do not predict
439 what a baseline scores on *held-out forms*, and the two diverge sharply. Across 12 corpora no zero-
440 parameter separability statistic we tried predicts untrained-encoder accuracy ($R^2 = -2.8$): the
441 statistic saturates while accuracy does not. We report this as a limitation of the tool rather than tuning
442 a statistic until it correlated. The operational consequence is step 2 of the checklist: run both baselines
443 rather than reading a projection off the screen.

444 Relatedly, the two baseline families measure different things and neither substitutes for the other.
445 On EQNET ONEVARPOLY13 the bag scores 0.15 where the untrained encoder scores 0.66; on
446 Feynman-10 the ordering reverses (1.00 versus 0.13). A recursive nonlinearity is sensitive to tree
447 shape a bag discards, and blind to count differences a bag sees.

448 I Per-Equation Heterogeneity

449 Identification accuracy on a K -equation suite is usually reported as a mean over seeds with a seed-
450 level interval. That answers “would a rerun give this number?” when the question of interest is
451 “would other equations give this number?” The two differ substantially here: per-equation leakage on
452 the 10-equation suite ranges from -0.05 (energy density) to 1.12 (wavenumber, whose held-out form
453 is a training duplicate: trivially solvable, so renaming destroys the only cue). Resampling equations
454 rather than seeds widens the interval by roughly $6\times$. We report equation-level bootstraps throughout
455 and recommend that any benchmark with fewer than a few dozen classes do the same; with $K=10$ a
456 seed-level interval is close to meaningless as a statement about the task.